

Executive Summary

This project focuses on performing **Exploratory Data Analysis (EDA)** on a student performance dataset to understand the key factors influencing academic outcomes in subjects like math, reading, and writing.

The primary objective of this analysis is to explore the dataset, identify patterns, uncover relationships between variables, and prepare the data for future machine learning applications.

Dataset Overview

The dataset consists of multiple features related to student demographics, family background, and study habits, including:

- Gender
- Ethnic Group
- Parental Education
- Lunch Type
- Test Preparation
- Weekly Study Hours
- Academic Scores (Math, Reading, Writing)

Data Cleaning and Preparation

During the analysis, the dataset was carefully examined and prepared for further exploration:

- Checked for **missing values and duplicates**
- Categorized variables into **categorical and numerical types**
- Prepared the dataset for visualization and modeling

Exploratory Data Analysis

Several visualizations were created to understand the distribution and relationships between variables

Distribution Analysis

- Academic scores (Math, Reading, Writing) show a **balanced distribution** with slight variations
- Most students fall within the **average performance range**, with few extreme values

Relationship Analysis

- Strong relationship observed between **Math, Reading, and Writing scores**
- Students performing well in one subject tend to perform well in others

Categorical Impact Analysis

- **Weekly Study Hours** show a positive impact on student performance
- **Parental Education** plays a significant role in influencing academic scores
- Performance patterns vary across **different demographic groups**

Detection

- Some extreme values were identified, representing **high-performing or low-performing students**

Key Insights

- Students who dedicate more time to studying generally achieve **higher scores**
- A strong correlation exists between all academic subjects
- Family background, especially **parental education**, influences performance
- The dataset contains meaningful patterns that can be leveraged for predictive modeling

Conclusion

This EDA project highlights how data analysis can uncover important insights about student performance. By exploring relationships between study habits and academic scores, we gain a deeper understanding of the factors that contribute to success.

The analysis not only provides valuable insights but also prepares the dataset for building reliable machine learning models.